$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

Substituting the points we get,

$$\frac{y+3}{-4+3} = \frac{x-5}{7-5}$$

$$\Rightarrow \quad 2y+6 = -x+5$$

Therefore, $\quad x+2y+1 = 0$

DO YOU KNOW?

The great mathematical physicists like Galileo and Newton used coordinate geometry to characterize the motions of objects in plane and space.

Example 5.24 Two buildings of different heights are located at opposite sides of each other. If a heavy rod is attached joining the terrace of the buildings from $(6,10)$ to $(14,12)$, find the equation of the rod joining the buildings ?

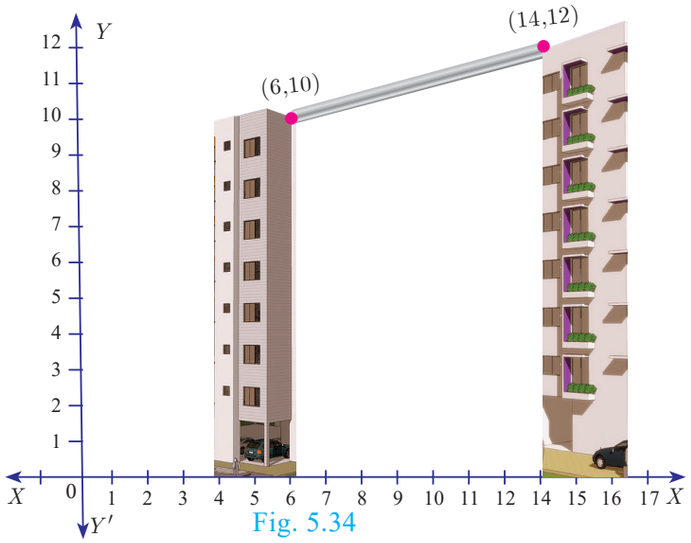


Fig. 5.34

Solution Let $A(6,10)$, $B(14,12)$ be the points denoting the terrace of the buildings.

The equation of the rod is the equation of the straight line passing through $A(6,10)$ and $B(14,12)$

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1} \quad \text{gives} \quad \frac{y-10}{12-10} = \frac{x-6}{14-6}$$

$$\frac{y-10}{2} = \frac{x-6}{8}$$

Therefore, $\quad x-4y+34 = 0$

Hence, equation of the rod is $x-4y+34=0$

5.5.7 Intercept Form

We will find the equation of a line whose intercepts are a and b on the coordinate axes respectively.

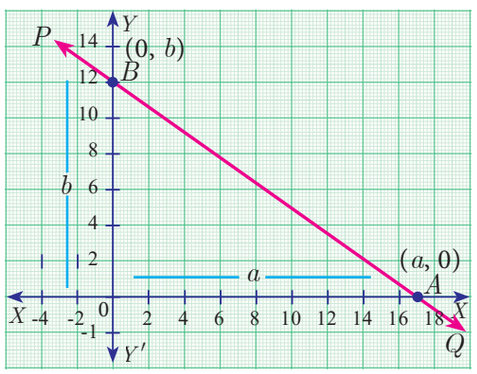


Fig. 5.35

Let PQ be a line meeting X axis at A and Y axis at B. Let $OA=a$, $OB=b$. Then the coordinates of A and B are $(a,0)$ and $(0,b)$ respectively. Therefore, the equation of the line joining A and B is

$$\frac{y-0}{b-0} = \frac{x-a}{0-a} \quad \text{we get,} \quad \frac{y}{b} = \frac{x-a}{-a} \quad \text{gives} \quad \frac{y}{b} = \frac{-x}{a}+1$$

Hence, $\frac{x}{a}+\frac{y}{b}=1$ (Intercept form of a line)

10th_Maths_Chapter 5_English.indd 225 12/27/2024 3:16:35 PM